

Agentic AI Digital Transformation Hackathon

Banking Challenge

Problem Statement

Governed Agentic AI for Banking Fraud Investigation and Response

Banks process large volumes of digital transactions and fraud alerts. Investigation often requires analysts to combine transaction behaviour, customer context, device/channel signals, fraud rules and other evidence before deciding whether a case should proceed, be verified, held or escalated. Your challenge is to redesign this process using governed Agentic AI while preserving appropriate human decision authority.

Overall Requirement: Design a governed Agentic AI solution that transforms the current manual/semi-manual process into a functional, integrated workflow with appropriate Human-in-the-Loop controls, open-model comparison, guardrails, expert validation, and a publishable project website.

Stage 1 - Structured Thinking, STOIC Prompt Engineering & Research (5 Marks)

Before building the solution, analyse the problem systematically using the thinking frameworks and STOIC Prompt Engineering discussed in class.

- Complete Critical Thinking and Systems Thinking analysis.
- Use STOIC Prompt Engineering for brainstorming and show prompt refinement.
- Generate and screen multiple solution ideas for usefulness, realism and feasibility.
- Conduct brief market and technology research on existing AI/automation solutions.
- Justify the final process and Agentic AI opportunity selected.

Stage 2 - Process Mapping & Agentic AI Architecture (5 Marks)

Prepare three clear diagrams showing the transformation from the current process to the proposed governed Agentic AI process.

- AS-IS Process Diagram: actors, activities, systems, decisions, handoffs, delays, bottlenecks and exceptions.
- TO-BE Agentic AI Process Diagram: orchestrator, specialised agents, tools, decisions, human approvals, escalations and outcomes.
- Technical & Governance Architecture Diagram: UI/UX, workflow/orchestration, agents/LLMs, tools/data, guardrails and human oversight.
- Mark TO-BE activities as A = Autonomous, H = Human-in-the-Loop, E = Escalation.

Stage 3 - Integrated Functional Agentic AI Solution (20 Marks)

Build one integrated working application. The UI/UX and the automation are parts of the same solution and must be connected.

- Use n8n or an equivalent tool for workflow automation/orchestration.
- Use Replit, Lovable, Streamlit, Flask, Cursor, Bolt or an equivalent platform for the UI/UX.
- Illustrative flow: Fraud Alert -> Orchestrator -> Transaction Analysis Agent -> Customer Behaviour Agent -> Risk/Policy Agent -> Recommendation Agent -> Human Fraud Investigator -> Outcome

- Demonstrate workflow triggering from the UI, multiple agent/logical roles, conditional routing, tool/data integration, HITL approval, escalation and exception handling.
- Test at least one normal case, one ambiguous/exception case and one high-risk case.
- Compare at least three open/open-weight models (for example Qwen, Llama, Mistral, Gemma, DeepSeek or approved equivalents) using the same cases, prompts, evidence and output format.
- Compare model performance using suitable metrics such as classification quality, escalation accuracy, false positives, hallucination, explanation quality, structured-output compliance, latency and task completion.
- Capture numbered and captioned screenshots of workflow, model runs, test cases and UI outputs.

Stage 4 - AI Guardrails & Governance/Compliance (5 Marks)

Prepare a concise guardrails draft and map it to a suitable AI, security or sector governance framework.

- Address relevant risks such as privacy, confidential data, hallucination, bias, prompt injection, unsafe autonomous action, access control, decision thresholds, human approval, logging and auditability.
- Select a suitable framework such as NIST AI RMF, ISO/IEC 42001, ISO/IEC 27001 or another approved sector-specific framework.
- Show the mapping as: Risk -> Guardrail -> Governance Requirement -> Implementation in the PoC.

Stage 5 - Industry Expert Interview & Final Validation (10 Marks)

The expert profile must be approved by the faculty before the assessed interview is conducted. The interview should preferably take place after the prototype is sufficiently developed so the expert can validate the proposed process and working solution.

- Submit the complete interview video and complete transcript.
- Submit the expert profile and evidence of professional designation.
- Show the expert the AS-IS process, TO-BE process, architecture, working PoC, HITL controls and proposed guardrails
- Document the expert's comments on feasibility, risks, usefulness and suggested improvements.
- You may ask follow-up questions in addition to the five mandatory questions below.

Five Mandatory Expert Questions

1. What makes fraud investigation a complex task in your organisation? What information, rules, evidence and judgement must an investigator normally consider?
2. Which people, systems, data sources or departments does a fraud investigator depend on, and where do these dependencies create difficulties or delays?
3. How important is speed in fraud investigation, and which stages of the process are most time-sensitive?
4. What types and volume of information must investigators examine, and what problems arise when information is incomplete or spread across different systems?
5. What kinds of unusual or ambiguous fraud cases are difficult to handle using standard rules, and how are such cases currently managed?

Industry Expert Marks Matrix

Component	Marks	Criteria
Completion of Expert Interview & Validation	5	Approved expert interviewed; interview completed; full video submitted; full transcript submitted; mandatory questions covered; feedback on the process and PoC documented.
Level of Industry Expert	5	Marks awarded according to the approved designation matrix below.

Expert Level	Indicative Titles	Marks
Director Level	Director, Senior Director, VP, AVP, Functional Head, Practice Head, Department Head, CXO or equivalent	5
Manager Level	Manager, Senior Manager, Program Manager, Project Manager, Operations Manager or equivalent domain manager	3
Analyst / Specialist Level	Analyst, Senior Analyst, Consultant, Senior Consultant, Specialist, Engineer or equivalent	2
Below Analyst / Not Relevant / Unapproved	Junior/entry-level profile, unrelated role, or profile not approved by faculty	0

Mandatory approval rule: Before scheduling the interview, submit the expert name, current designation, organisation, relevant function/domain, available experience details, professional profile/LinkedIn link, and a short justification. The faculty-approved designation level will determine the marks. If the expert changes, fresh approval is required.

Stage 6 - Final Report, Publishable Website & Demonstration Video (5 Marks)

Submit a concise project report, a public-facing project website, and a 10-15 minute final demonstration video.

- Report: problem, STOIC/research, AS-IS/TO-BE diagrams, architecture, workflow screenshots, model comparison, test results, guardrails/governance, expert findings, limitations and recommendations.
- Publishable website: problem statement, process diagrams, architecture, PoC/demo, model comparison, benchmark results, guardrails, expert validation summary, limitations and final video.
- Transcript of the expert interview with question and answer.
- Every team member must participate meaningfully in the final demonstration video.

Evaluation Matrix

Stage	Component	Marks	Criteria
Stage 1	Structured Thinking, STOIC Prompt Engineering & Research	5	Critical Thinking; Systems Thinking; STOIC Prompt Engineering; Prompt Refinement; Idea Screening; Market Research; Technology Research; Existing Solution Review; Opportunity Justification
Stage 2	Process Mapping & Agentic AI Architecture	5	AS-IS Process; TO-BE Agentic Process; Agent Roles; Human-in-the-Loop Mapping; Escalation Mapping; Technical Architecture; Governance Architecture
Stage 3	Integrated Functional Agentic AI Solution	20	Functional UI/UX; n8n/Equivalent Automation; UI-Workflow Integration; Agent Orchestration; Conditional Logic; Tool/Data Integration; HITL; Escalation; Exception Handling; Test Cases; Open-Model Comparison; Comparative Performance Analysis
Stage 4	AI Guardrails & Governance/Compliance	5	AI Risk Identification; Data Guardrails; Model Guardrails; Action Guardrails; Human Oversight; Auditability; Governance Framework Selection; Compliance Mapping
Stage 5	Industry Expert Interview & Validation	10	Faculty Profile Approval; Expert Level; Domain Relevance; Five Mandatory Questions; Interview Video; Transcript; Expert Validation; Feasibility/Risk/Improvement Feedback
Stage 6	Final Demonstration Video	5	Problem Explanation; Research; Process Diagrams; Live Agentic AI Demonstration; UI/UX Demo; Model Comparison; Governance; Expert Findings; Team Participation
	TOTAL	50	